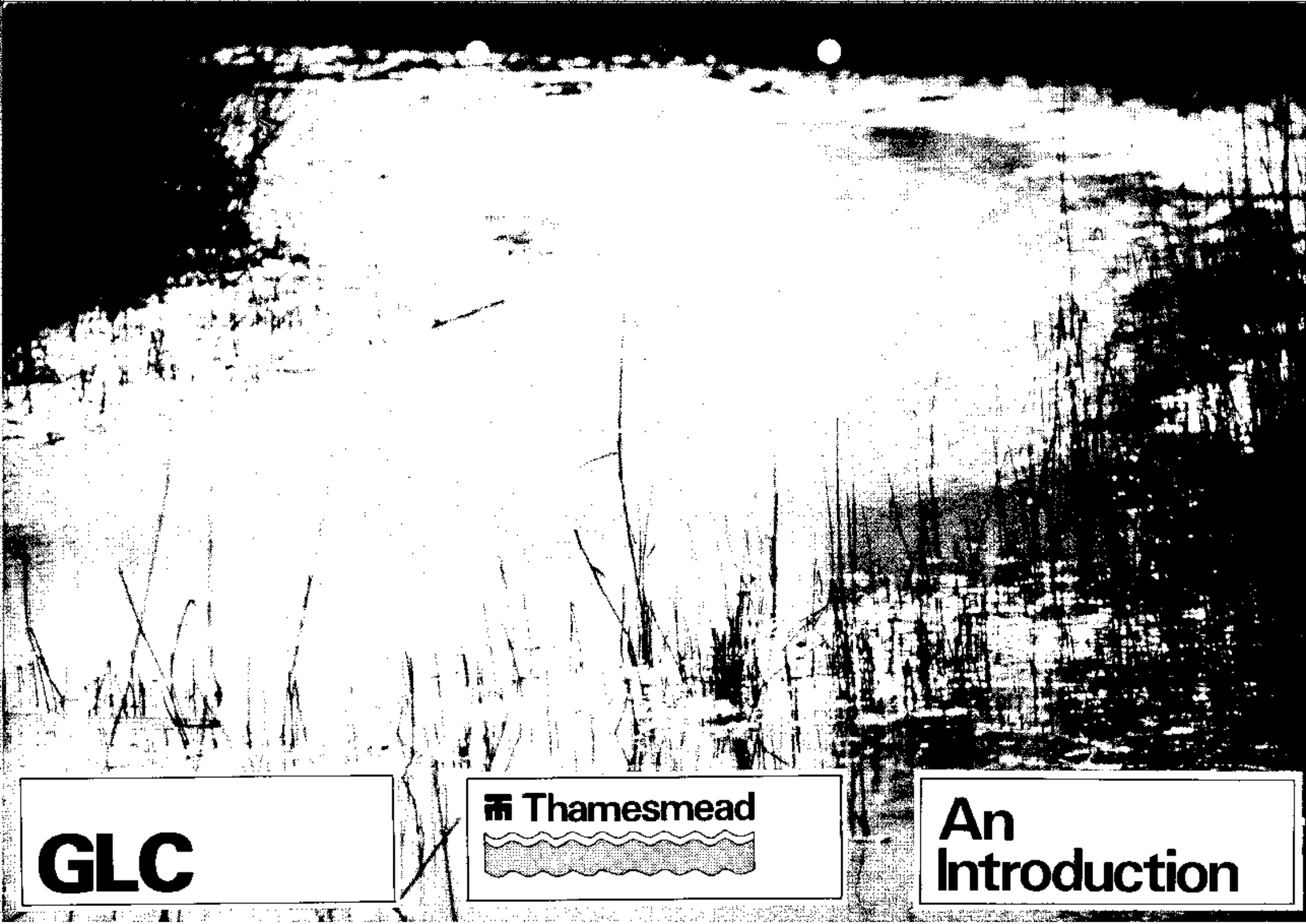


GLC

Thamesmead

An Introduction

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This brochure tells the background story to Thamesmead before development began.

THE SITE

A 5.6 kilometre (3.5 mile) stretch of marshy hinterland on the south side of the River Thames situated on the boundary between the London Boroughs of Bexley and Greenwich, the total area being 648 hectares (1600 acres).

BRIEF HISTORY

The original river bank was probably built up by the Romans. There is evidence of a small Roman settlement at nearby Plumstead.

In the twelfth century Sir Richard de Lucy built Lesnes Abbey to the south as a foundation for Augustinian Canons. The monks played their part in the reclaiming and draining of the land. The Abbey was demolished in 1524. However, in 1930, the old London County Council (LCC) and other authorities started re-excavating and restoring the remaining ruins.

Between 1530 and 1600 the marsh lands suffered damaging floods and no satisfactory reclamation was carried out. Responsibility was eventually assigned to Commissioners of Sewers who raised a local tax called 'wall-scot' to cover the expense of maintaining the river wall and draining the land

Meanwhile, Henry VIII established Woolwich Dockyard just up the river from Thamesmead. The land downstream which included the site of Thamesmead became known as The Warren. In the seventeenth century the area became a naval and military centre. Guns were set up in The Warren and large storehouses built for naval guns. Premises designed by Sir John Vanbrugh for a brass foundry were built in 1716, the first time a manufacturing industry was established in the area. The Royal Artillery built its first barracks at Tower Place, Woolwich in 1719 to be near their source of supply. The Royal Military Academy grew from a small academy in Tower Place and a purpose built hall for the cadets was provided on The Warren. The cadets and their tutors had a reputation for drunkenness and corruption and caused many local scandals. In 1840 fifty of them fought with bludgeons at Charlton Horn Fair. The school eventually was moved to Sandhurst.

The first royal visit to The Warren was by George III in 1772. He watched the casting of brass guns, a 42 pounder being bored and examined various pieces of artillery. On his second visit in 1805, he re-christened the area The Arsenal. It was too war-like a place, he thought, to be named The Warren, a place for timid creatures such as rabbits.

Before Australia was colonised, convicts were put to hard labour in the area. They lived in old rotting hulks on the river. One of their jobs was to fill up swampy ground with mud and debris from excavations at St. Katherine's Dock near the Tower of London. Their pay was threepence a day. Until 1817 convicts were buried at The Warren and, after that, in a cemetery on the Essex bank of the river.

The nineteenth century was a period for expansion for the Arsenal. By 1890, it covered 323.7 hectares (800 acres) and included practice and experimental ranges. The War Office added 161.8 hectares (400 acres) of marshland where the Arsenal built a group of "danger buildings" and a pier on the river bank linked to the main railway by a branch line. At this time, eleven moated magazines were also built.

However, from the end of the First World War the Arsenal was begining to be criticised, it lay too far from sources of coal and steel supplies. It was difficult to get people

Ruins of Lesnes Abbey

to work there. It was badly laid out and provided an easy target for air attack. This latter warning proved to be justified in the 1939-45 war when a thousand men were killed or injured in air raids. By the early fifties, the Ministry of Defence admitted that the Arsenal was inefficient and, by then, under used. Substantial lands were sold to the London County Council for the Woolwich Industrial Estate. The transaction resulted in a surplus of about 20.2 hectares (50 acres) which now forms part of Thamesmead.

Subsequently, the Ministry agreed to release all their surplus lands to the LCC - a total of 404.6 hectares (1000 acres). To this the LCC added 151.8 hectares (375 acres) from the environs of the Sewage Treatment Works at Crossness. (These works on the north eastern periphery of Thamesmead were built in the Italian style in 1865 and opportunities occasionally arise for the public to view the pumping engines which are a fine example of Victorian engineering.)

By the early sixties, therefore, the area offered about 648 hectares (1600 acres) ready for development.



The marshes in the late eighteenth century. Part of the topographical map of Kent by Andrews Drury and Herbert, 1769.

THE NEED FOR HOUSING

From the outset it was felt that the area made available for development should be used to help meet London's most urgent need - housing.

At the end of the 1939-45 war London faced an acute housing shortage. Temporary factory-made homes were provided on vacant sites but these could house only a few of the many thousands who were homeless. The old London County Council looked outside its boundaries for a solution. Cottage estates were built in outer London for 100,000 people while a redevelopment programme for inner London was being implemented. In addition, the Government introduced legislation whereby eight new towns could be created in the home counties to help alleviate London's problem. This was reinforced in 1952 by the Town Development Act which encouraged the expansion of existing towns by agreement between local authorities.

Although both measures would have enabled many Londoners to have homes, it was realised that the pace of development would be very slow. By the mid-fifties the LCC was seeking Government approval to build, at top speed, its own new town for 100,000 people near Hook in Hampshire. The proposal, however, had to be abandoned because of the strenuous opposition to the scheme by the Hampshire County Council.

It was in this situation that the Ministry of Housing and Local Government began to urge the release of Government and railway surplus lands. This resulted in the sites of old airports becoming available (Hendon and Croydon) also Kidbrooke Depot and Royal Victoria Yard, Deptford, lands which have now been developed with housing. In 1963 the Government estimated a need of at least 500,000 homes in the following ten years and hinted in a White Paper that not only the airport lands could be provided but specifically "further development may be possible at Woolwich Arsenal".

In the early sixties, the LCC were developing ideas for building on the 151.8 hectares (375 acres) of land which had originally been reserved for the possible expansion of Crossness Sewage Treatment Works. Other sites had been added and the final area was 202.3 hectares (500 acres) between Abbey Wood and Belvedere stations. It was soon realised that development would be difficult. This was the first confrontation with problems which were to beset architects later when designing for Thamesmead.

The area lay well below the river's high water level and there was an ever-present danger of flood. The water table was high. Ditches overflowed and rainwater would lie on top of the ground for days. (One of the local councils even had a bye-law in force stipulating that the floor level of any habitable room should be at least 2.4 metres (8 feet) above ordnance datum.) The soil was an unstable mixture of peat and clay in uneven layers over a gravel bed. Access to the area could be difficult and the lands were remote from the main lines of communication.

Of the many schemes considered, one imaginative example was for island villages on decks 3.6 metres (12 feet) above the ground. Each island contained 31-storey tower blocks with low rise blocks. The whole complex included a lake, schools and local shopping centres and was knit together with a system of canals and roads. The suggested method of construction was to be industrialised building using standard components manufactured by a purpose-built factory.

In fact, many of the ideas behind this proposal were used eventually at Thamesmead. However, this particular scheme had to be abandoned as it was not big enough to provide future employment; there was no scope for a shopping centre of any size and communications were inadequate.

Nevertheless, the idea for a "little town on stilts" stimulated great interest. The possibilities for developing a large marsh site were there and as pressure built up for action to deal with the housing shortage so the fences round the Arsenal lands began to come down.



THE "MASTER PLAN"

A team of GLC planners, architects and valuers assembled their plans and ideas for the development of the area into a "Master Plan" which was submitted to the then Ministry of Housing and Local Government for approval.

The principle aims of this tentative scheme were:

a) To provide for a community of around 60,000 persons;
b) To provide sufficient employment to ensure that the increase in the number of commuters to central London would be kept within reasonable bounds;
c) To build a Central Area of appropriate size for such a community;
d) To form open space and transport links with surrounding areas;
e) To develop fully the riverside situation of the site and exploit its watery nature;
f) To organise circulation within the community so that pedestrians enjoyed many traffic-free areas.

Considerable thought had been given to creating a good environment and it was felt that this could best be achieved by developing at an overall density of 247 persons per hectare (100 persons per acre) which would give the necessary homes and amenities required by a population of 60,000. From this could be determined the scale of schools, open spaces, welfare requirements, shops, churches and public houses which would be needed. The density figure allowed flexibility: houses with gardens at 173 persons per hectare (70 ppa) up to urban type flats and maisonettes at 346 persons per hectare (140 ppa).

It was appreciated that Council housing tended to group people of a similar mode and style of living, usually those at the lower end of the economic scale. As far as possible the development was to be a microcosm of society and it was decided to encourage private developers to build 35% of the housing for those able to purchase their own homes.

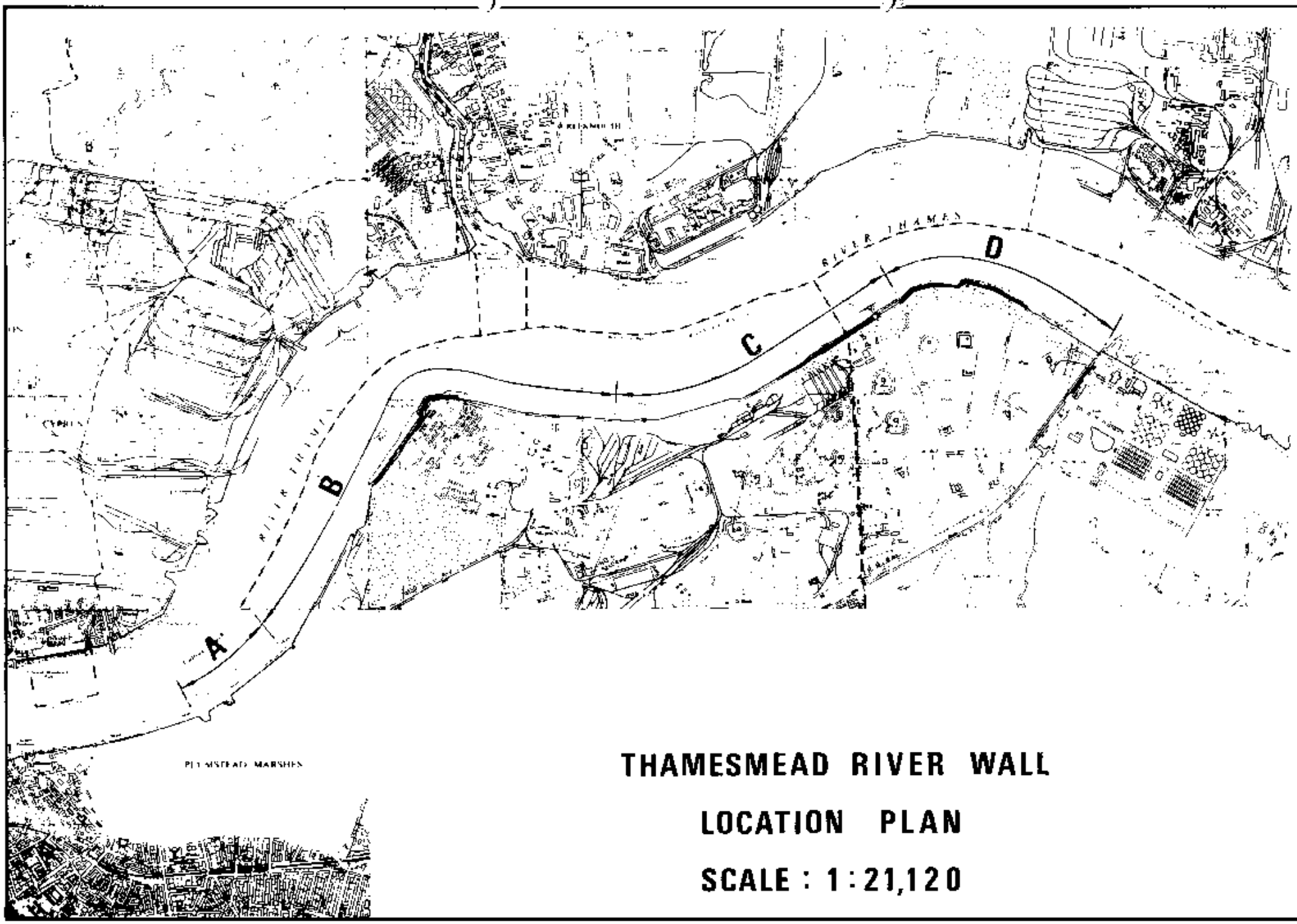
Early models and drawings of Thamesmead showed a pattern emerging: tall buildings along the river bank with a spur running north/south from north of Abbey Wood Station to the town centre roughly midway along the river front where there would be shopping, commerce, places of entertainment alongside a marina. Moreover, development against major roads would be linear in form, buildings up in density towards the future Central Area and forming thereby a protective wind and noise barrier between these roads and the low rise housing areas.

Development would be in stages, a series of related environmental areas each having its own primary schools, local shops, churches, public houses, and some local sources of employment.

It should always be borne in mind that the "Master Plan" was never intended to be a definitive design of the area. It was realised that, as work progressed, needs would change, fresh ideas would emerge, certain goals would prove to be untenable and unforseen difficulties would have to be solved requiring innovation. In this sense, the scheme has evolved and continues to develop by rational processes.

A major re-appraisal was carried out in 1972. It was realised that private housing development was unlikely to reach densities much higher than 123 persons per hectare (50 ppa). However, it was imperative to the success of the community that the aim for a variety of tenure was maintained. As a result, it was decided to reduce the future overall population figure to around 45,000 to 48,000. In turn, this has necessitated various adjustments to the concommitant needs of the community.

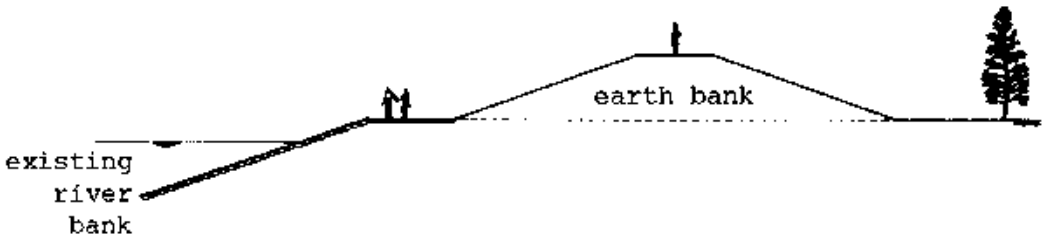
Model of the 1967 Master Plan



THAMESMEAD RIVER WALL

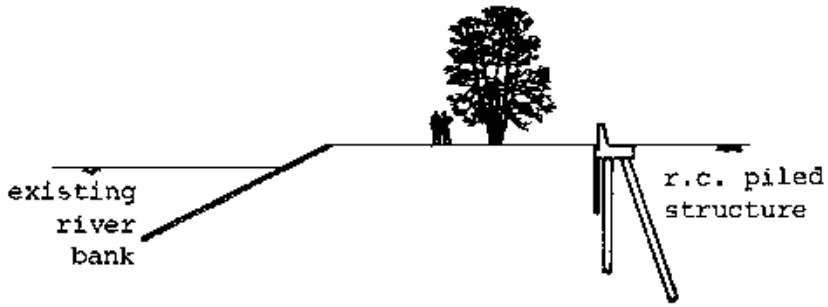
LOCATION PLAN

SCALE : 1 : 21,120



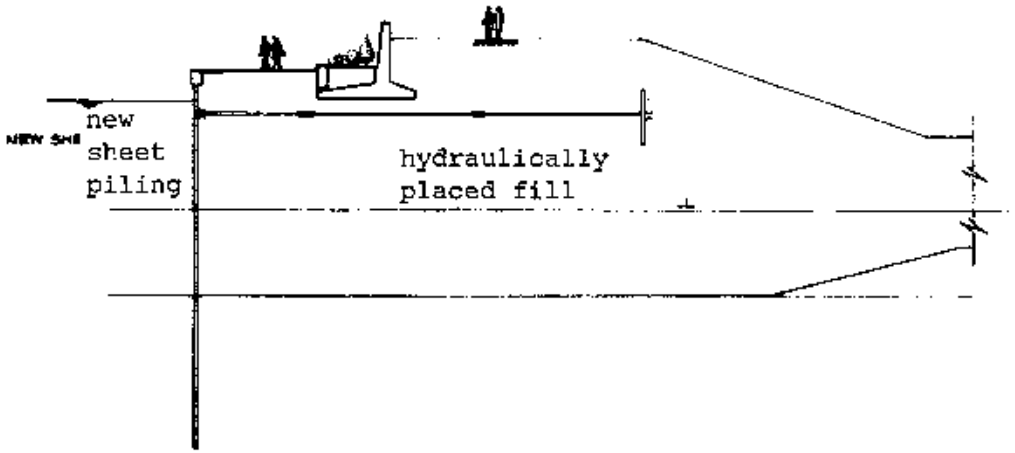
TYPICAL SECTION LENGTHS B & D

FIG 4



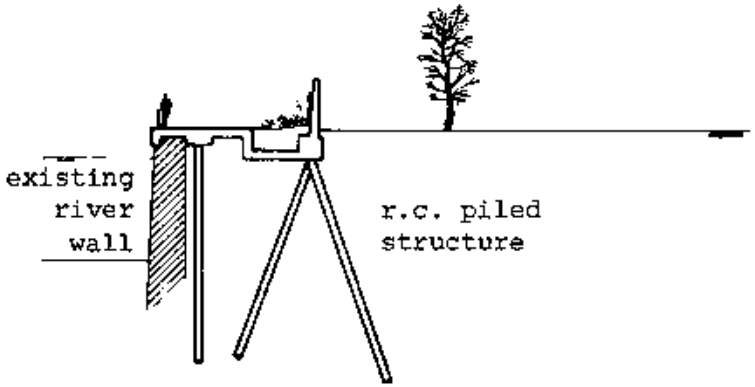
TYPICAL SECTION LENGTH B

FIG 2



TYPICAL SECTION LENGTH C

FIG 3



TYPICAL SECTION LENGTH A

FIG 1

THAMESMEAD RIVER WALL TYPICAL SECTIONS

THROUGH NEW WORK

SOLVING THE PROBLEMS

Flood Danger The Thamesmead Architect and the Director of Public Health Engineering have developed designs which will not only provide defence against tidal flood levels but also a continuous riverside walkway of 5.2 kilometres (3.25 miles) with opportunities for shrub planting and landscaping. The extent of the work consists mainly in raising the existing river banks to a level of 7.2 metres, that is 1.8 metres above the existing levels. The frontage conveniently falls into four areas, each requiring its own special treatment (see diagrams on pages 7 and 8):

a) Old Woolwich Arsenal quay wall. A reinforced concrete retaining wall was found necessary, set back 8 metres from the existing wall and founded on raked bearing piles. The surface between the old river wall and the flood wall to be sealed to prevent seepage of water.

b) From the Old Arsenal quay wall to Abbey Sluice. A mixture of low earthen banks set back to where the old river bank is massive with vertical concrete flood retaining walls elsewhere.

c) Abbey Sluice to a point 150 metres of Crossness Jetty. Here the new flood defence works are combined with the reclamation of land. A steel sheet piled retaining wall has been constructed in the river with a full flood defence concrete wall to the landward side. The cost of the work was about £3.5 million.

d) Eastern end of the reclamation area to Crossness Sewage Treatment Works. Here the flood defence works take the form of an additional earth embankment set back from the existing river bank.

Surface Water Drainage Historically the surface water run-off from the hills flowed naturally across Thamesmead by way of a network of dykes and ditches, discharging by gravity into the river through three sluices. Because of the difference in level between the river and the site, discharge was possible for only a short period on either side of low tide.

This system is being replaced by five lakes and an interconnecting network of canals totalling 32.3 hectares (80 acres) of water area. A major pumping station is being constructed to pump all stormwater that would raise the level of the water storage area by more than 160 millimetres (6 inches) and to pump against any tide level.

Ground Conditions All buildings and many of the civil engineering structures are being constructed on piled foundations to transmit the weight of the structure through the clay, peat and alluvium to the load bearing gravel stratum below. The piles at Thamesmead are designed to carry loads of between 30.4 and 60.9 tonnes (30 and 60 tons).

Roads and Sewers Because of the difficult ground conditions, stability of roads on embankments presented a special problem. After trying various methods to pump out or absorb the water content to ensure settlement, it was eventually decided to weight the embankments down with surplus rubble for a period before final road works were done.

The bottoms of trenches for deep sewers were pushed up by artesian pressure from the gravel below. The usual method of relieving such pressure is by pumping, but in this case there was a danger of causing subsidence to buildings and foundations. Steel sheet piles for the trenches some 2.4 metres (8 feet) into the gravel were tried and the gravel contained in the piles was injected with silicone soda for waterproofing. Although effective the method proved expensive and it was decided that sewers should henceforth be no more than at a depth of 3.04 metres (10 feet), relying on eight or nine pumping stations to lift sewage to main outfall sewers draining to Crossness Sewage Treatment Works.

PREPARATION FOR BUILDING

Although the "Master Plan" was not approved until 1967, there was much to be done before any development could take place.

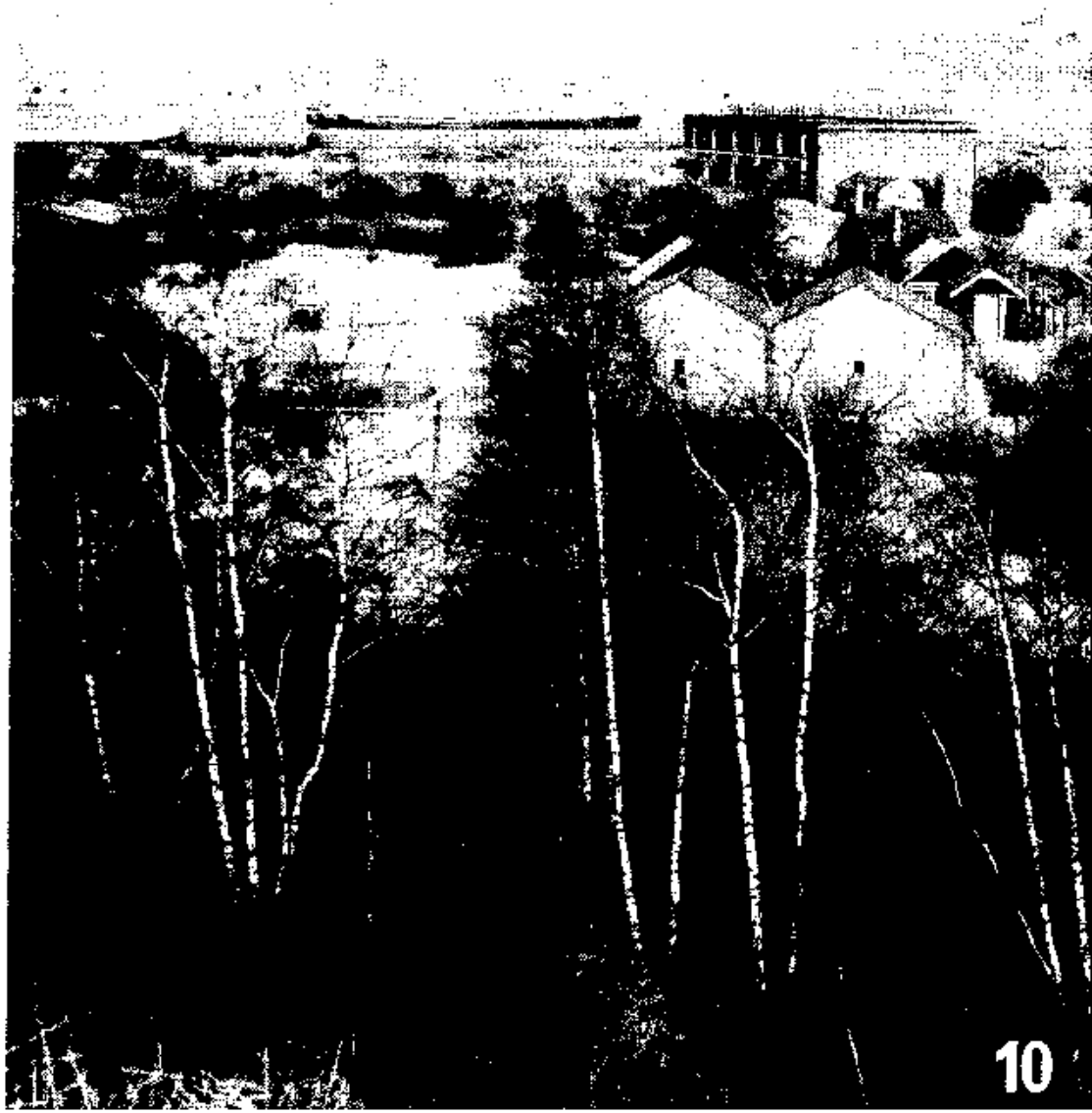
Purchase of Lands The Arsenal lands were crossed by old ditches, roads and rail tracks and contained a large number of old or partly derelict buildings. These were major factors in respect of the price to be paid to the Ministry of Defence. It was also necessary to make assumptions about the land uses proposed, the conditions for building, particularly the abnormal foundation costs envisaged, and the lengthy time scale for the new development. After much discussion between the Ministry and the Council final agreement was reached resulting in a price of £6,820,000 for 414.8 hectares (1025 acres). The agreement covered the release of the lands in stages and the first area of 166.7 hectares (412 acres) was handed over on 1 January 1968. Further handovers in 1970 and 1972 completed the physical assembly of the site while the Council met the purchase price in a series of phased payments.

Survey It was essential after acquisition to get an accurate picture of the site. A detailed survey was begun in 1967 which was to take three years to complete because of the size of the site, the arduous terrain and because frequent spells of bad weather often held up progress. The discovery of some old unexploded missiles further hampered the work of the Council's Land Surveyors.

The survey now forms the basis of all new development proposals. Survey work continues in providing extra detail in specific areas and for the plotting of new development as it proceeds.

Clearance and Demolition Old Arsenal buildings were demolished as soon as they were handed over as part of a parcel of land. However structures of interest and merit have been given a reprieve. For example, the so-called

Typical Old Arsenal buildings awaiting demolition

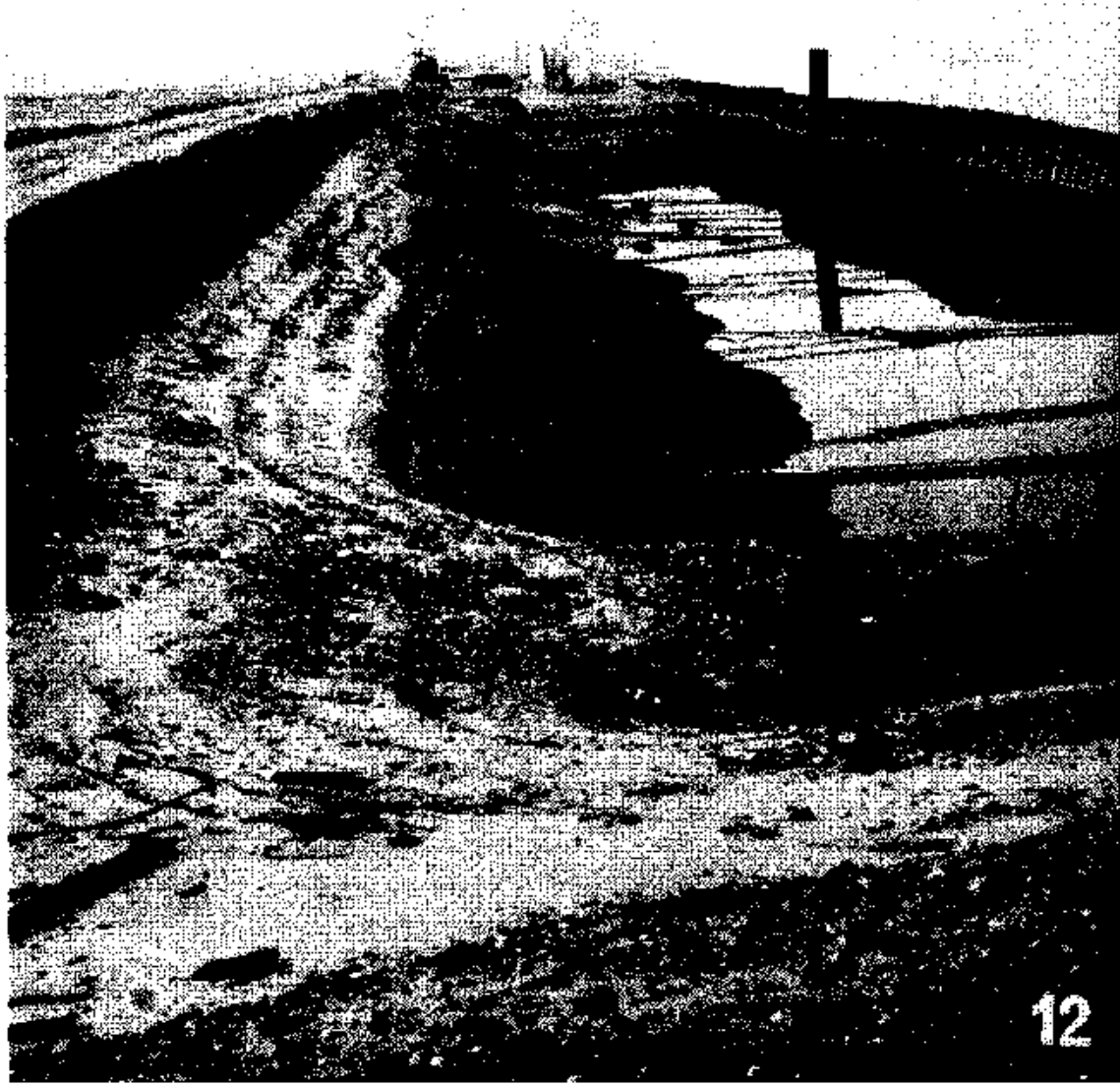




"tumps". These were previously small shell filling factories surrounded by moats to prevent the spread of fire in case of an accident. Several of these moats have been incorporated into the new system of canals.

Demolition work was on a large scale and not always an easy matter. One of the structures included a 101.6 tonnes (100 ton) gantry crane on its roof and pits 21 metres (70 feet) deep. It had been used for producing guns for the Navy's "Ironclads". The building has now gone and the pits filled in, awaiting development as part of Area III.

Landscaping Tree belts were planted around Crossness Sewage Treatment Works and the new Eastern Industrial Estate before the first houses were built. The GLC's Parks Department are constantly a few steps ahead of the builders, healing the scars left by buildings that have been demolished and seeing that it is not too long before any bareness in the new development is relieved with greenery.



Aerial view of the site - circa 1945

Land Reclamation

ADMINISTRATION

As Thamesmead lies partly in Bexley and partly in Greenwich, both these London Borough Councils are responsible for many of its services. The main development, however, is being undertaken by the Greater London Council. The Thamesmead Committee of the GLC is responsible for the planning, construction and management of the development. The Committee's resources are determined annually by the Council in relation to the needs of the Greater London area as a whole. Twenty-two members of the Council sit on the Committee and six additional members are nominated by the London Boroughs of Bexley and Greenwich. Two further members are co-opted from outside the Council, one being nominated by the Thamesmead Community Association. The GLC staff who advise the Committee and carry out its policies are drawn from various Council departments. The Department of Architecture and Civic Design is responsible for the design and construction of the housing and other buildings and also for civil engineering work. The Thamesmead Manager is responsible for co-ordinating administration within the Council and for liaising with the Borough Councils and the Inner London Education Authority. The Director-General is the Council's principal adviser, responsible for co-ordination of policy and administration, including clerking the Thamesmead Committee, and for publicity through the Public Information Branch. The Housing Department is the principal client department and advises on all housing matters; development, management and maintenance of dwellings.

Other departments who play a part are:

The Department of Mechanical and Electrical Engineering
The Parks Department (landscaping and open spaces)
Department of Planning and Transportation (highways, strategic and environmental planning)
Treasurers Department (financial policy, annual budget etc)
Valuation and Estates Department (property implications, property management other than housing)

Education Services at Thamesmead are provided by the Inner London Education Authority (ILEA) for the western part within the London Borough of Greenwich and Bexley Council for the remainder of the site.

A range of social services - old peoples needs, libraries, recreational facilities etc - are provided by Greenwich and Bexley Councils.

FINANCIAL ASPECTS

It was estimated that the net capital cost of Thamesmead would be about £200,000,000 and the Councils investment would form the greater part with smaller contributions by the Greenwich and Bexley Borough Councils, the Inner London Education Authority and by the Government through grants and subsidies. Total capital expenditure from the start of the scheme to 31 March 1976 was about £80.9 M.

The bulk of local authority money has to be borrowed and repaid with interest of the lifetime of the asset acquired or built - normally 60 years for housing. Interest charges are met from revenue income, that is rents charged to private developers for sites or buildings, council house rents and from local rates.

Southmere Lake



THE THAMESMEAD BONUS

Although the difficultics of the site for building purposes cannot be overstressed, the natural advantages of a marshy site should not be overlooked. Wild shrubland of thorn, sloe, silver birch; clusters of sycamore, planes, poplars; wild grasses, flowers, reeds, marsh plants - much is being done to conserve some of these delights. Ducks and marsh birds are still to be seen in the area, as well as kestrels and the occasional rarer bird. In fact, a nature trail has been formed.

Moreover, the nature of the site has much to offer the boating enthusiast. Since the first lake, Southmere, was opened for boating in 1971, small dinghies have tacked up and down on fine days and rowing boats have been hired out to those wishing to mingle with the swans. The system of canals to be provided will make it possible to travel across the whole site by boat in due course, a distance of over 7 kilometres (4.5 miles).

FURTHER INFORMATION

A detailed bibliography (Research Bibliography No. 62 June 1975) has been compiled by the GLC Research Library and is available from the GLC Bookshop, The County Hall, London, SE1 7PB, price 40 pence.

Other brochures in this series are:

Thamesmead - The First Areas
Thamesmead - Future Developments

available from the address below.

All enquiries about the work of the GLC Department of Architecture and Civic Design should be addressed to:

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